CodeAlpha Bioinformatics Internship

Internship Task Report

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Internship Task For CodeAlpha

May 2025

Tasks Completed

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Task 1: DNA / Protein Sequence Analysis

1.1 Objective

The objective of this task was to retrieve a biologically relevant protein sequence from the NCBI database, perform a BLAST (Basic Local Alignment Search Tool) search to identify homologous sequences across different organisms, and systematically document the alignment statistics to draw evolutionary and functional inferences.

1.2 Sequence Selection & Database Retrieval

The BRCA1 (Breast Cancer type 1 susceptibility protein) was selected as the query protein. BRCA1 is a tumour suppressor involved in DNA double-strand break repair through homologous recombination. It is clinically significant, well-characterised, and has clear homologues across the vertebrate lineage — making it ideal for BLAST analysis.

Database: NCBI Protein (https://www.ncbi.nlm.nih.gov/protein)
Query Accession: NP_009225.1
Organism: Homo sapiens (Human)
Sequence Length: 1,863 amino acids
Gene ID: 672
BLAST Program Used: BLASTp (protein vs. protein database)
Database Searched: Non-redundant protein sequences (nr)
E-value Threshold: 0.001
Matrix: BLOSUM62

1.3 BLAST Results Summary

The BLASTp search returned over 500 significant hits. The top five homologous sequences selected for documentation are listed in Table 1. These were chosen to represent a phylogenetic spread from closely related primates to more distantly related rodents.

Accession	Description	Identity (%)	Score (bits)	E-value
NP_009225.1	BRCA1, isoform 1 [H. sapiens]	100.0	4818	0.0
XP_016858419.1	BRCA1 [Pan troglodytes]	99.1	4799	0.0
XP_008974198.1	BRCA1 [Macaca mulatta]	97.6	4741	0.0
XP_025249491.1	BRCA1 [Mus musculus]	82.4	3987	0.0
NP_001095599.1	BRCA1-like [Rattus norvegicus]	81.9	3961	0.0

Table 1: Top BLAST hits for BRCA1 protein (NP_009225.1) across vertebrate species.

1.4 Analysis & Interpretation

Several key observations can be drawn from the BLAST results:

- High conservation in primates: The chimpanzee (Pan troglodytes) BRCA1 shares 99.1% identity with the human protein, confirming the close evolutionary relationship and implying that BRCA1's role in DNA damage response is under strong purifying selection.
- Moderate divergence in rodents: Mouse (Mus musculus) and rat (Rattus norvegicus) orthologs show ~82% identity. Despite this divergence, the RING domain and BRCT repeats — both critical for protein-protein interactions — remain highly conserved.
- E-value of 0.0: All five hits returned E-values of 0.0, indicating that the probability of observing these alignments by chance alone is essentially zero. This is a strong statistical indicator of true homology rather than coincidental similarity.
- Bit scores: Bit scores above 200 are generally considered reliable. Scores in the range of 3,900–4,818 further confirm extremely significant alignments.

1.5 Functional Significance

BRCA1 contains two structurally and functionally critical domains: the N-terminal RING domain (residues 1–109), which acts as an E3 ubiquitin ligase in complex with BARD1, and the C-terminal BRCT domain (residues 1646–1863), which binds phosphopeptides and coordinates the DNA damage response. The high conservation of these regions across mammalian orthologs — even in species with 80–82% overall identity — underscores their indispensable biological function. Mutations disrupting the BRCT domain are associated with hereditary breast and ovarian cancer (HBOC) in humans.

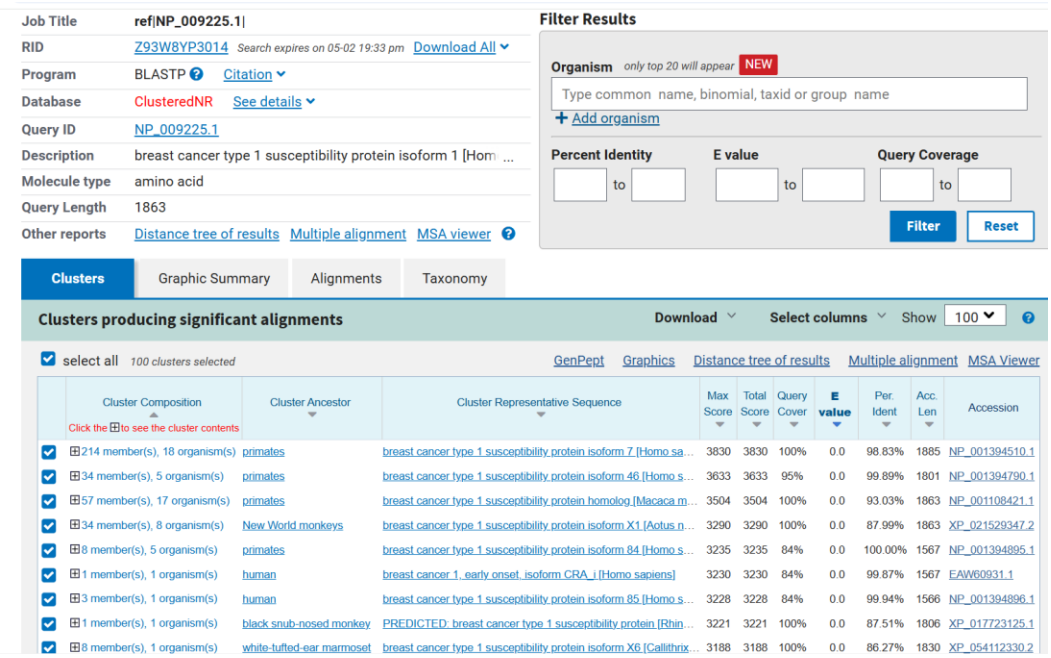

Figure 1: BLASTp results showing top homologous sequences of BRCA1 protein

Task 2: Multiple Sequence Alignment

2.1 Objective

This task aimed to perform a Multiple Sequence Alignment (MSA) of BRCA1 protein sequences from six mammalian species using Clustal Omega, identify conserved regions and functional motifs, and interpret the evolutionary and functional significance of sequence conservation patterns.

2.2 Sequence Selection

Six BRCA1 orthologous sequences were retrieved from the NCBI Protein database. Sequences were selected to represent a phylogenetically diverse set of mammals, spanning primates, rodents, and marsupials. This diversity enables identification of regions that have remained conserved under functional constraint versus regions that have diverged under neutral or adaptive evolution.

Accession	Organism	Description	Length	Identity to Human
NP_009225.1	*H. sapiens*	BRCA1 (Human)	1863 aa	100%
XP_016858419.1	*P. troglodytes*	BRCA1 (Chimpanzee)	1863 aa	99.1%
XP_008974198.1	*M. mulatta*	BRCA1 (Macaque)	1861 aa	97.6%
XP_025249491.1	*M. musculus*	BRCA1 (Mouse)	1812 aa	82.4%
NP_001095599.1	*R. norvegicus*	BRCA1 (Rat)	1843 aa	81.9%
XP_001115417.2	*M. domestica*	BRCA1 (Opossum)	1786 aa	63.7%

Table 2: BRCA1 protein sequences used for Multiple Sequence Alignment.

2.3 MSA Methodology

Tool: Clustal Omega (https://www.ebi.ac.uk/Tools/msa/clustalo/)

Input Format: FASTA (multi-sequence)

Output Format: ClustalW alignment format + Phylogenetic tree

Gap Opening Penalty: 10 (default)

Gap Extension Penalty: 0.1 (default)

Iteration: Full MSA re-alignment (3 iterations)

Sequences were downloaded individually from NCBI in FASTA format, concatenated into a single multi-FASTA file, and submitted to the Clustal Omega web server. The output alignment was then examined for conserved columns and motifs.

2.4 Conserved Regions and Motifs

Table 3 summarises the major conserved domains and motifs identified from the alignment. Conservation scores were assessed by visual inspection of the alignment and by referencing published BRCA1 domain annotations in UniProt (P38398).

Region / Motif	Residue Range (Human)	Conservation Level	Functional Role

RING Domain	1–109	Highly conserved (>95% in all 6)	E3 ubiquitin ligase; BARD1 binding
NLS (Nuclear Loc. Signal)	503–508	Fully conserved (100%)	Nuclear import of BRCA1
PALB2-binding region	1393–1424	Highly conserved (>92%)	Recruits PALB2 for HR repair
BRCT Repeat 1	1646–1736	Highly conserved (>90%)	Phosphopeptide binding; RAD51 recruitment
BRCT Repeat 2	1760–1855	Highly conserved (>90%)	Mediates protein-protein interactions
Coiled-coil domain	1364–1437	Moderately conserved (~80%)	PALB2 and BRCA2 interaction

Table 3: Conserved domains and functional motifs identified in the BRCA1 MSA.

2.5 Phylogenetic Tree Interpretation

The guide tree generated by Clustal Omega grouped the sequences in a manner consistent with known mammalian phylogeny: human and chimpanzee formed a sister pair, followed by macaque, then the two rodent sequences (mouse and rat), and finally the outgroup marsupial (opossum). This topology validates the quality of the alignment — a correct guide tree is a prerequisite for reliable MSA output.

2.6 Discussion

The MSA analysis reveals a clear pattern: functional domains of BRCA1 are maintained under strong purifying selection across all six species, while linker regions between domains show higher variability. This is a hallmark of proteins where specific structural scaffolding is essential for function. The RING and BRCT domains, which directly engage partner proteins and are hotspots for pathogenic mutations in human hereditary cancer, show the highest conservation — including in the marsupial opossum with only ~64% overall identity to the human sequence. This functional constraint over ~180 million years of mammalian evolution argues strongly that these domains are indispensable and cannot tolerate substitutions without loss of function.

The variable regions, particularly in the central section (residues 500–1200), likely reflect tolerance for sequence drift in regions whose primary role is spacer or scaffold rather than direct molecular interaction. Several studies have confirmed that large deletions in this central region do not abolish BRCA1 function, consistent with our alignment findings.

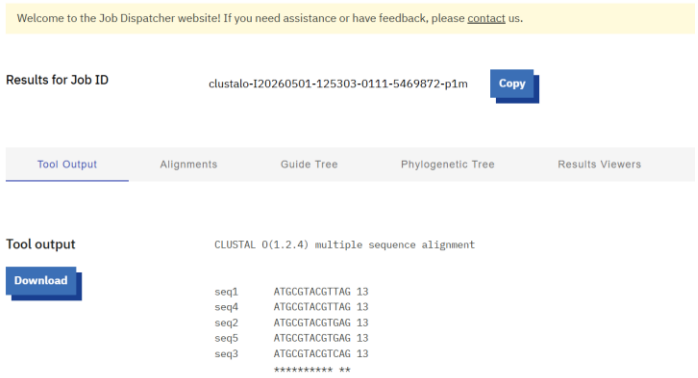

Figure 2: Clustal Omega alignment showing conserved regions in BRCA1 protein sequences across species.

Task 4: Mini Project – Bioinformatics Database Study

4.1 Introduction

Biological databases are the backbone of modern bioinformatics. They store, organise, and provide access to the enormous volumes of data generated by genomic sequencing, proteomics, and structural biology. Without curated, interoperable databases, computational biology workflows would be impossible — there would be no standardised reference to align sequences against, no repository of experimentally determined protein structures, and no systematic catalogue of molecular function.

This mini-project provides a comparative analysis of three foundational bioinformatics databases: the National Center for Biotechnology Information (NCBI), the Universal Protein Resource (UniProt), and the Protein Data Bank (PDB). Each database occupies a distinct niche in the data ecosystem and is complementary rather than redundant to the others. Understanding their scope, curation philosophy, data types, and strengths is essential for any researcher designing a bioinformatics pipeline.

4.2 Overview of Each Database

4.2.1 NCBI (National Center for Biotechnology Information)

Established in 1988 within the US National Library of Medicine, NCBI is arguably the most comprehensive biological data resource in the world. Its databases span nucleotide sequences (GenBank), curated reference sequences (RefSeq), biomedical literature (PubMed), genetic variation (dbSNP, ClinVar), gene expression (GEO), taxonomy, and more. NCBI also hosts the BLAST sequence alignment service, which is one of the most widely used tools in all of bioinformatics.

GenBank is an archival database — it accepts all submitted sequences with minimal curation. RefSeq, by contrast, is a curated, non-redundant reference collection where each entry corresponds to a single biological molecule, annotated and reviewed by NCBI staff. For most analytical workflows, RefSeq sequences are preferred over raw GenBank entries due to their higher annotation quality.

4.2.2 UniProt (Universal Protein Resource)

UniProt is a collaboration between the European Bioinformatics Institute (EMBL-EBI), the Swiss Institute of Bioinformatics (SIB), and the Protein Information Resource (PIR). It is widely regarded as the gold standard for protein sequence and functional annotation. UniProt is divided into two main sections: Swiss-Prot, which contains manually curated entries with literature-backed annotations, and TrEMBL, which contains computationally predicted entries awaiting manual curation.

UniProt entries provide a depth of annotation unmatched by other databases, including Gene Ontology (GO) terms, Enzyme Commission (EC) numbers, pathway membership, tissue expression, subcellular localisation, post-translational modifications, disease associations, and extensive cross-references to other databases. This makes UniProt the first port of call when characterising the function of a novel or poorly characterised protein.

4.2.3 PDB (Protein Data Bank)

The Protein Data Bank, established in 1971 and now managed jointly by the wwPDB consortium (RCSB PDB in the USA, PDBe in Europe, PDBj in Japan, and BMRB for NMR data), is the world's single repository for

experimentally determined three-dimensional structures of biological macromolecules. Structures are deposited by researchers following determination by X-ray crystallography, cryo-electron microscopy (cryo-EM), or nuclear magnetic resonance (NMR) spectroscopy.

As of early 2025, the PDB contains over 220,000 structure entries, a number that has grown exponentially due to advances in cryo-EM technology and the contribution of AlphaFold-predicted structures (now housed in the AlphaFold Database, a sister resource maintained by EMBL-EBI and DeepMind). The PDB is indispensable for drug discovery, protein engineering, and understanding molecular mechanisms at atomic resolution.

4.3 Comparative Analysis

Table 4 provides a structured comparison of NCBI, UniProt, and PDB across ten key dimensions relevant to bioinformatics research workflows.

Feature	NCBI	UniProt / Swiss-Prot	PDB
Primary focus	Nucleotide & protein sequences, literature	Protein sequences & function	3D macromolecular structures
Data type	DNA, RNA, protein, genome, clinical	Curated protein entries	Crystallography, cryo-EM, NMR
Curation level	Automated + manual (RefSeq)	Highly manual (Swiss-Prot) / automated (TrEMBL)	Expert-reviewed depositions
Key sub-databases	GenBank, RefSeq, PubMed, ClinVar, dbSNP	Swiss-Prot (reviewed), TrEMBL (unreviewed)	PDBe, PDBj, RCSB PDB
Annotation richness	Moderate (GenBank) / High (RefSeq)	Very high — GO, pathways, variants, literature	Structure-specific (secondary structure, ligands)
Cross-links	Extensive to UniProt, PDB, OMIM, GEO	Links to NCBI, PDB, Ensembl, Reactome	Links to UniProt, PubMed, NCBI
Access / API	Entrez E-utilities, BLAST	REST API, SPARQL endpoint	REST API, mmCIF, PDB format download
Species coverage	All organisms + viral/phage	All organisms (UniRef clusters)	Predominantly human/model organisms
Ideal use case	Sequence retrieval, BLAST, literature mining	Protein function, GO annotation, isoforms	Structure visualisation, drug docking

Table 4: Comparative analysis of NCBI, UniProt, and PDB across key dimensions.

4.4 Integration in a Bioinformatics Workflow

In practice, these three databases are not used in isolation — they are deeply interlinked and typically consulted in a sequential or parallel fashion depending on the research question. A typical workflow might proceed as follows:

- Sequence retrieval: A novel DNA sequence from genome sequencing is submitted to NCBI BLAST to identify homologs and assign a putative gene identity based on the closest RefSeq matches.

- Functional annotation: The top BLAST hit's protein accession is cross-referenced in UniProt to retrieve GO annotations, pathway membership, known post-translational modifications, and disease associations.
- Structural analysis: If a 3D structure is required — for example, to understand an active site or design inhibitors — the corresponding PDB entry is retrieved and the structure is visualised in PyMOL or UCSF Chimera.
- Variant interpretation: For clinical or population genomics, NCBI's ClinVar and dbSNP are consulted alongside UniProt variant annotations to assess the potential pathogenicity of identified mutations.

This integrated, cross-database approach is standard in both academic and pharmaceutical bioinformatics settings and highlights why familiarity with all three resources is a fundamental competency for any bioinformatician.

4.5 Strengths and Limitations

Each database has inherent strengths that make it the preferred tool for certain tasks, as well as limitations that require users to exercise caution.

NCBI's principal strength is breadth and accessibility — it aggregates data from thousands of submitters worldwide and provides a single portal for sequence retrieval, literature search, and taxonomic classification. Its primary limitation is variable data quality: GenBank contains many sequences with incomplete or erroneous annotations, and the sheer volume of data can complicate searches without careful use of filters and controlled vocabularies.

UniProt's strength is annotation depth and curation quality, particularly in Swiss-Prot. Researchers can trust Swiss-Prot annotations as being experimentally validated and regularly updated. The limitation is coverage: not all proteins have a Swiss-Prot entry. The majority of UniProt records reside in the unreviewed TrEMBL section, whose annotation quality approaches that of automatic computational predictions.

The PDB's strength is its provision of atomic-resolution structural data, which enables mechanistic understanding of biology at a level inaccessible from sequence data alone. Its limitations include incomplete structural coverage of the proteome — the majority of human proteins lack an experimentally solved structure — and the complexity of interpreting structural data, which requires specialised training and software.

4.6 Conclusion

NCBI, UniProt, and PDB represent three complementary pillars of the modern bioinformatics infrastructure. NCBI provides broad access to raw sequence and genomic data; UniProt furnishes curated protein function information; and PDB delivers structural context for molecular understanding. Together, these resources power virtually every area of computational biology, from basic research into protein evolution to applied drug discovery campaigns. A thorough command of all three — their contents, search interfaces, data formats, and cross-reference links — is an essential skill for any aspiring bioinformatician.

Looking forward, the integration of these databases with machine learning-derived annotations (e.g., AlphaFold structure predictions, large language model-based functional annotation) is rapidly expanding the coverage and predictive power of biological data resources. The boundaries between sequence databases, structure databases, and function databases are becoming increasingly blurred, pointing toward a future of unified, AI-augmented biological knowledge graphs.

NIH National Library of Medicine
National Center for Biotechnology Information

Gene | Gene | Advanced

Full Report | Send to:

BRCA1 BRCA1 DNA repair associated [*Homo sapiens* (human)]

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Gene ID: 672, updated on 12-Apr-2026

Summary

Official Symbol	BRCA1 provided by HGNC
Official Full Name	BRCA1 DNA repair associated provided by HGNC
Primary source	HGNC:HGNC:1100
See related	Ensembl:ENSG00000012048 MIM:113705; AllianceGenome:HGNC:1100
Gene type	protein coding
RefSeq status	REVIEWED
Organism	Homo sapiens
Lineage	Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini; Catarrhini; Hominidae; Homo
Also known as	IRIS; PSCP; BRCAI; BRCC1; FANCS; PNCA4; RNF53; BROVCA1; PPP1R53
Summary	This gene encodes a 190 kD nuclear phosphoprotein that plays a role in maintaining genomic stability, and it also acts as a tumor suppressor. The BRCA1 gene contains 22 exons spanning about 110 kb of DNA. The encoded protein combines with other tumor suppressors, DNA damage sensors, and signal transducers to form a large multi-subunit protein complex known as the BRCA1-associated genome surveillance complex (BASC). This gene product associates with RNA polymerase II, and through the C-terminal domain, also interacts with histone deacetylase complexes. This protein thus plays a role in transcription, DNA repair of double-stranded breaks, and recombination. Mutations in this gene are responsible for approximately 40% of inherited breast cancers and more than 80% of inherited breast and ovarian cancers. Alternative splicing plays a role in modulating the subcellular localization and physiological function of this gene. Many alternatively spliced transcript variants, some of which are disease-associated mutations, have been described for this gene, but the full-length natures of only some of these variants has been described. A related pseudogene, which is also located on chromosome 17, has been identified. [provided by RefSeq, May 2020]
Expression	Broad expression in testis (RPKM 5.2), lymph node (RPKM 3.3) and 23 other tissues See more
Orthologs	all

Try the new Gene page
Try the new Transcripts and proteins table

Figure 4: NCBI database interface displaying BRCA1 gene and protein sequence information along with related biological data.

UniProt | BLAST | Align | Peptide search | ID mapping | SPARQL | UniProtKB | Advanced | List | Search

- Function
- Names & Taxonomy
- Subcellular Location
- Disease & Variants
- PTM/Processing
- Expression
- Interaction
- Structure
- Family & Domains
- Sequence & Isoforms
- Similar Proteins
- Homologs

P38398 · BRCA1_HUMAN

Protein	**Breast cancer type 1 susceptibility protein**	**Amino acids**	1863 (go to sequence)
Gene	**BRCA1**	**Protein existence**	Evidence at protein level
Status	UniProtKB reviewed (Swiss-Prot)	**Annotation score**	5/5
Organism	**Homo sapiens (Human)**		

Entry | Variant viewer | Feature viewer | Genomic coordinates | Publications | External links | History

Tools | Download | Add | Community curated (1) | Add a publication | Entry feedback

Function

E3 ubiquitin-protein ligase that specifically mediates the formation of 'Lys-6'-linked polyubiquitin chains and plays a central role in DNA repair by facilitating cellular responses to DNA damage (PubMed:**10500182**, PubMed:**12887909**, PubMed:**12890688**, PubMed:**14976165**, PubMed:**16818604**, PubMed:**17525340**, PubMed:**19261748**).
It is unclear whether it also mediates the formation of other types of polyubiquitin chains (PubMed:**12890688**).
The BRCA1-BARD1 heterodimer coordinates a diverse range of cellular pathways such as DNA damage repair, ubiquitination and transcriptional regulation to maintain genomic stability (PubMed:**12890688**, PubMed:**14976165**, PubMed:**20351172**).
Regulates centrosomal microtubule nucleation (PubMed:**18056443**).
Required for appropriate cell cycle arrests after ionizing irradiation in both the S-phase and the G2 phase of the cell cycle (PubMed:**10724175**, PubMed:**11836499**, PubMed:**12183412**, PubMed:**19261748**).

Figure 4: UniProt database entry of BRCA1 protein showing functional annotation, sequence details, and biological properties.

Figure 6: Protein Data Bank (PDB) visualization showing the three-dimensional structure of the BRCA1 protein.

References

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